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Prof. Kenneth M. Merz, Jr., Editor-in-Chief
*Journal of Chemical Information and Modeling*
American Chemical Society

Dear Prof. Merz,

We submit our manuscript **"Target breadth and mutation resilience in African-natural-product-inspired antimalarial chemotypes: a computational analysis"** for consideration as an Article in *Journal of Chemical Information and Modeling*.

This work addresses a critical methodological challenge in computational antimalarial drug discovery: evaluating target breadth and mutation resilience as distinct, complementary properties when prioritizing candidates from African natural-product-inspired chemical space, while maintaining clear boundaries between computational evidence and biological activity.

We analyze a 17-member polypharmacology-oriented cohort across four mechanistically diverse antimalarial targets (PfDHFR, PfCRT, PfClpP, and PfATP4) using target-specific structural anchors and a per-target resistance-resilience scoring (RRS) framework. The workflow integrates three evidence layers: chemical-space expansion (65 856 hybrid library molecules $\rightarrow$ 19 913 synthesizable candidates), target-anchored AutoDock Vina docking generating a complete $17 \times 4$ scoring matrix (68/68 geometric quality pass rate), and per-target RRS calculated separately for PfDHFR (N51I, C59R, S108N, I164L) and PfCRT (K76T, K76A) mutant panels. The cohort yielded six A*, five B, five C, and one D RRS profiles (range 68.2–111.7), with weak cross-metric associations after multiplicity correction (PNS–RRS $\rho = -0.559$, ACSI–RRS $\rho = -0.132$).

**Key Enhancements:**

This submission incorporates comprehensive validation, physicochemical characterization, and methodological documentation:

1. **Validation Documentation:** Transparent assessment through three complementary approaches: (a) external benchmark (DEKOIS 2.0, PfDHFR ROC-AUC 0.45, reported honestly as near-chance), (b) enrichment validation (MMV Malaria Box, ROC-AUC 0.924–1.000), and (c) redocking validation (5 ligands, RMSD $< 2.0$ Å). Complete metrics in Supporting Information Section S12.

2. **Physicochemical Characterization:** All 17 candidates exhibit drug-like properties (100 % Lipinski/Veber compliance, mean QED 0.703, low predicted ADMET risk). Details in Supporting Information Tables S4–S6.

3. **Multi-Parameter Optimization Framework:** Candidate selection utilized a seven-component MPO framework (Vina affinity 35%, DiffDock confidence 25%, QED 20%, ADMET 15%, Lipinski penalty 5%, polypharmacology bonus) with MPO $\geq 0.40$ threshold. Complete mathematical formulation and weight rationale in Supporting Information Section S10.

4. **Methodological Rigor:** Complete grid box specifications for all four targets, computational efficiency documentation (99.3 % cost reduction via centroid-based library reduction), and strict evidence boundaries maintained throughout.

**Evidence Boundaries:**

The study maintains clear distinctions between computational evidence layers. Results do not claim experimental affinity, confirmed simultaneous target engagement, biological resistance circumvention, or measured $IC_{50}/EC_{50}$ values. The workflow produces: (a) four-target docking profiles from which polypharmacology hypotheses can be generated, (b) per-target RRS values summarizing mutation-panel docking ratios, and (c) exploratory cross-metric associations connecting distinct evidence types without conflation.

**Accessibility and Impact:**

The centroid-based library reduction strategy addresses a critical accessibility gap, particularly for research groups in endemic regions (94 % of malaria cases). By reducing computational costs by orders of magnitude while maintaining structural diversity, this workflow enables systematic multi-target screening without high-performance infrastructure, supporting reproducibility and broader participation in computational antimalarial research.

**Reproducibility:**

Complete computational provenance is provided: source code, candidate manifests, receptor/ligand files, grid configurations, raw docking poses, geometric quality filters, RRS protocols, and machine-readable tables are archived at `https://github.com/NanaEngo/Malaria_codesV2`. Supporting Information (17 pages, 12 sections, 9 tables, 2 figures) includes complete validation datasets, physicochemical characterization, and methodological specifications.

This integrated approach, connecting chemical-space exploration to target-anchored docking and mutation-aware scoring while maintaining strict evidence boundaries, provides a reproducible template for hypothesis generation in resistance-challenged therapeutic areas. The work will interest the JCIM readership engaged in computational drug discovery, multi-target prioritization strategies, African natural product chemoinformatics, and validation frameworks for docking-derived hypotheses.

This manuscript is original, not under consideration elsewhere, and all authors have reviewed and approved the submission. We declare no competing financial interests.

Sincerely,

**Myke Vital Sao Temgoua**
(on behalf of all co-authors)